

Haloalkanes & Haloarenes

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Haloalkanes (R-X) are formed by replacing a hydrogen of an alkane with a halogen; haloarenes (Ar-X) have the halogen directly bonded to an aromatic ring.
- SN2 (substitution nucleophilic bimolecular) mechanism: single-step, backside attack by nucleophile, causing inversion of configuration; favored for primary/methyl halides.
- SN1 (substitution nucleophilic unimolecular) mechanism: two-step, via a carbocation intermediate, can lead to racemization; favored for tertiary halides (stable carbocation).
- Rate of SN1 depends only on [substrate] ($\text{rate} = k[\text{RX}]$); rate of SN2 depends on both [substrate] and [nucleophile] ($\text{rate} = k[\text{RX}][\text{Nu}^-]$).
- Haloarenes are less reactive toward nucleophilic substitution than haloalkanes because the C-X bond has partial double bond character (due to resonance with the ring) and is shorter/stronger.
- Elimination reactions (E1, E2) compete with substitution; E2 is a one-step bimolecular elimination (base removes a beta-H as the leaving group departs), favored by strong bases and bulky substrates.
- Grignard reagents (R-MgX) are formed from haloalkanes and magnesium in dry ether; they are powerful nucleophiles used to form new C-C bonds.



SN2: backside attack, X leaves

SN2 mechanism: a nucleophile attacks from the backside as the leaving group departs.

2. Formula Sheet

SN2 rate law	Rate = k [RX][Nu-]
SN1 rate law	Rate = k [RX]
Grignard reagent formation	R-X + Mg (dry ether) → R-MgX
Wurtz reaction	2 R-X + 2 Na → R-R + 2 NaX
Finkelstein reaction	R-X + NaI (acetone) → R-I + NaX
SN1 carbocation stability order	3 degree > 2 degree > 1 degree
E2 elimination (general)	R-CH ₂ -CH ₂ -X + base → R-CH=CH ₂ + HX
Dipole moment order (C-X bond)	C-F > C-Cl > C-Br > C-I (bond polarity decreases down the group)

3. Practice Problems (22 questions)

1. State whether a primary alkyl halide favors SN1 or SN2 mechanism, and explain why.
2. State whether a tertiary alkyl halide favors SN1 or SN2 mechanism, and explain why.
3. Explain why SN2 reactions proceed with inversion of configuration (Walden inversion).
4. State the rate law for an SN1 reaction and explain what it implies about the rate-determining step.
5. Explain why haloarenes are less reactive than haloalkanes toward nucleophilic substitution.
6. State the reagents and conditions required to prepare a Grignard reagent from an alkyl halide.
7. Explain the Wurtz reaction and write a general equation for it.
8. State the difference between E1 and E2 elimination mechanisms.
9. Explain why SN1 reactions can lead to racemization of the product, while SN2 reactions lead to inversion.
10. State the Finkelstein reaction and explain why acetone is used as the solvent.
11. Explain why bulky nucleophiles/bases favor elimination (E2) over substitution (SN2).
12. Rank the following in order of increasing bond strength: C-F, C-I, C-Br, C-Cl.
13. State why tertiary alkyl halides do not readily undergo SN2 reactions.
14. Explain how a Grignard reagent reacts with water, and why Grignard reactions must be carried out in anhydrous conditions.
15. State the general trend in boiling points of haloalkanes as the halogen changes from F to I (same alkyl group).
16. Explain why the C-X bond in haloarenes is shorter than in haloalkanes.
17. State one major use of chloroform (CHCl₃) or a similar haloalkane.
18. Explain why polar aprotic solvents favor SN2 reactions over polar protic solvents.

19. State the effect of increasing solvent polarity on the rate of an SN1 reaction.
20. Explain why haloalkanes are generally insoluble in water but soluble in organic solvents.
21. Write the product of the reaction of ethyl bromide ($\text{CH}_3\text{CH}_2\text{Br}$) with sodium hydroxide (aqueous, SN2 conditions).
22. State why the presence of a good leaving group (e.g., I- vs F-) affects the rate of both SN1 and SN2 reactions.
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Answer Key

1. SN2 - primary carbocations are unstable, so a concerted backside-attack mechanism is favored (no carbocation intermediate needed)
2. SN1 - tertiary carbocation is highly stabilized (hyperconjugation + inductive effect), favoring a stepwise ionization mechanism
3. Conceptual - nucleophile attacks from the side opposite the leaving group, flipping the configuration like an umbrella in the wind
4. Conceptual - rate = $k[\text{RX}]$ implies the rate-determining step involves only the substrate (formation of the carbocation), independent of nucleophile concentration
5. Conceptual - haloarene C-X bond has partial double bond character from resonance with the ring, making it shorter, stronger, and less reactive to substitution
6. Conceptual - alkyl halide + Mg metal in dry (anhydrous) ether
7. Conceptual - 2 alkyl halides react with sodium metal to form a longer-chain alkane (coupling reaction); $2\text{R-X} + 2\text{Na} \rightarrow \text{R-R} + 2\text{NaX}$
8. Conceptual - E1 is two-step (via carbocation, unimolecular rate-determining step); E2 is one-step (concerted, bimolecular, base removes H as leaving group departs)
9. Conceptual - SN1 forms a planar carbocation intermediate attacked from either face equally, giving a racemic mixture; SN2 is a single concerted inversion step, giving one specific stereochemical outcome
10. Conceptual - Finkelstein reaction converts R-Cl/R-Br to R-I using NaI in acetone; acetone dissolves NaI but not the NaCl/NaBr byproduct, driving the reaction forward (precipitation)
11. Conceptual - bulky bases have difficulty approaching the carbon for backside attack (SN2), so they preferentially abstract a beta-hydrogen instead (E2)
12. $\text{C-I} < \text{C-Br} < \text{C-Cl} < \text{C-F}$ (increasing bond strength)
13. Conceptual - steric hindrance from three bulky alkyl groups blocks backside nucleophilic attack required for SN2
14. Conceptual - Grignard reagents react violently with water ($\text{R-MgX} + \text{H}_2\text{O} \rightarrow \text{RH} + \text{Mg(OH)X}$), so anhydrous conditions are essential to prevent this side reaction
15. Conceptual - boiling point generally increases from F to I due to increasing molecular weight and stronger van der Waals forces
16. Conceptual - resonance donation of ring pi electrons into the C-X bond gives it partial double-bond character, shortening it
17. Conceptual - e.g., chloroform historically used as an anesthetic, now mainly as a solvent
18. Conceptual - polar aprotic solvents solvate cations well but not anions, leaving the nucleophile more 'naked' and reactive for SN2 attack
19. Conceptual - increasing solvent polarity stabilizes the carbocation intermediate, increasing the rate of SN1 reactions
20. Conceptual - haloalkanes are largely non-polar hydrocarbon-like molecules that cannot hydrogen-bond with water, but dissolve well in non-polar organic solvents

21. Product: ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) + NaBr (via $\text{S}_\text{N}2$ substitution)

22. Conceptual - better leaving groups (weaker C-X bond, more stable leaving anion) depart more easily, increasing the rate of both $\text{S}_\text{N}1$ (ionization step) and $\text{S}_\text{N}2$ (concerted departure) reactions

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.